Data-Backed Journal-Style Replication of *Beyond Local Search*

Replication Paper Project

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Abstract

This paper presents a local calculated replication of Beyond Local Search: Boundary-Spanning, Exploration, and Impact in the Optical Disk Industry using only processed datasets and result files available in the workspace. The replication preserves the source paper's core empirical logic while making every local substitution explicit. The current local batch contains 23,844 observations across 184 entities, and the batch headline estimate is -0.0389 for boundary_existing_field vs local with p = 0.3868. The local analogue supports the most novel boundary-spanning channel much more clearly than the moderate boundary-spanning channel. The contribution is a journal-style replication manuscript that separates direct source claims from executable local evidence.

Keywords: boundary-spanning search; exploration; inventive impact; patent classes; local calculated analogue

1 Introduction

This replication begins from a paper-first reading of Beyond Local Search: Boundary-Spanning, Exploration, and Impact in the Optical Disk Industry and treats the source article as an empirical argument that has to be rebuilt before it can be judged. The motivating question is: The source paper asks whether boundary-spanning search improves inventive impact relative to local search. The study is worth revisiting because The replication matters because boundary-spanning search claims are often treated broadly, while the original paper distinguishes among qualitatively different

forms of exploratory search. The original article does not rest on a descriptive regularity alone. It ties a theory about search, differentiation, timing, or organizational learning to a specific unit of analysis, sample frame, and estimand. A careful replication therefore has to recover the structure of that logic before any local coefficients are interpreted. The executable evidence base used in this manuscript contains 23,844 observations across 184 entities and spans 1986 to 2016 in the local executable panel.. Those numbers are not filler. They define the actual inferential scope of the local reconstruction and set the boundary on what can be claimed honestly from the available datasets. The paper's contribution is a transparent three-way search taxonomy on a local executable panel that separates local, boundary-existing-field, and boundary-new-field search. The paper is written to read like a formal journal article rather than an internal workflow memo. The main text develops the theoretical logic, data reconstruction, derivations, empirical design, and results in continuous prose, while the appendices preserve the audit trail required for reruns and later extensions.

2 Original Study, Theory, and Replication Boundary

The focal source article is [1]. The original study can be summarized as a mechanism paper with a sharp empirical core. The source paper contrasts local search with boundary-spanning exploration. Impact rather than patenting volume is the main empirical outcome. Industry-specific coding is part of the original design. That combination matters because the source article asks readers to believe not merely that two variables move together, but that a particular organizational or strategic mechanism explains why the relationship should appear in the data. In a setting like this, faithful replication requires more than reusing a title and rerunning a regression. The theoretical contribution lives in the way constructs are defined, how they enter the estimand, and which comparison the model is supposed to adjudicate. For that reason, this manuscript first reconstructs the published theory and only then turns to the local executable analogue. The replication boundary is therefore explicit. Direct source facts are carried over when they can be read from the published article itself. Local replication decisions appear only when the current workspace must translate the source design into verified processed datasets. The goal is disciplined mechanism recovery, not rhetorical imitation that quietly changes the design.

2.1 Direct Source Facts Versus Replication Decisions

The manuscript distinguishes direct source facts from local execution decisions. The source article establishes that The source paper emphasizes different kinds of boundary-spanning search. The local reconstruction therefore adopts the following bounded decision: The local panel reconstructs a three-way search taxonomy from experience counts. This bridge is treated as a visible design choice, not as a hidden substitution. The reason for spelling out this bridge in prose is that replication credibility depends on showing exactly where the manuscript is following the article and where it is making an explicit local translation. That is especially important in this project, because the current paper is built from processed datasets already present in the workspace rather than from a reconstructed raw archival pipeline. The manuscript distinguishes direct source facts from local execution decisions. The source article establishes that Impact is the key outcome. The local reconstruction therefore adopts the following bounded decision: The outcome remains impact-focused through citations and a top-tail indicator. This bridge is treated as a visible design choice, not as a hidden substitution. The reason for spelling out this bridge in prose is that replication credibility depends on showing exactly where the manuscript is following the article and where it is making an explicit local translation. That is especially important in this project, because the current paper is built from processed datasets already present in the workspace rather than from a reconstructed raw archival pipeline. The manuscript distinguishes direct source facts from local execution decisions. The source article establishes that Industry-specific coding is part of the original design. The local reconstruction therefore adopts the following bounded decision: The industry-specific coding is generalized to the processed multi-firm patent panel. This bridge is treated as a visible design choice, not as a hidden substitution. The reason for spelling out this bridge in prose is that replication credibility depends on showing exactly where the manuscript is following the article and where it is making an explicit local translation. That is especially important in this project, because the current paper is built from processed datasets already present in the workspace rather than from a reconstructed raw archival pipeline.

DirectSourceFact	LocalReplicationDecision
The source paper emphasizes different kinds of boundary-spanning search.	The local panel reconstructs a three-way search taxonomy from experience counts.
Impact is the key outcome.	The outcome remains impact-focused through citations and a top-tail indicator.
Industry-specific coding is part of the original design.	The industry-specific coding is generalized to the processed multi-firm patent panel.

Direct source facts and explicit local execution decisions carried in the body of the paper.

3 Data, Sample, and Measures

The local evidence chain is deliberately modular. The source paper is treated as the authority for the research question, theoretical mechanism, and published design. Estimation, by contrast, proceeds only through the verified local files. The required dataset types in this paper are Patent-level recombination panel built from the cleaned reproduction dataset. Search type reconstructed from prior organizational and field experience counts. Impact outcomes measured with patent-level inventive impact and a top-5-percent indicator. Operationally, the reconstruction moves through the following file chain: Analysis panel is backed by the local file rosenkopf_2001_analysis_dataset.csv; Coefficient export is backed by the local file rosenkopf_2001_model_coefficients.csv; Sample summary is backed by the local file rosenkopf_2001_sample_summary.csv. The resulting executable panel supplies 23,844 observations across 184 entities and covers 1986 to 2016 in the local executable panel.. Every numeric claim in the results section is intended to trace back to one of those

verified files rather than to a hand-copied number from the article. This data-lineage discipline matters for two reasons. First, it prevents the manuscript from smuggling unverified preprocessing steps into the design. Second, it forces every substitution to be visible. If a construct cannot be reproduced from the current processed datasets, the paper states the analogue directly instead of pretending that the original archive has been fully recreated. The local analogue reconstructs local, boundary-existing-field, and boundary-new-field search from organizational and field experience counts already materialized in the processed recombination panel. The local sample frame is Patent-level recombination panel derived from the cleaned local reproduction dataset. The observed time window in the executable panel is 1986 to 2016 in the local executable panel. The outcome is Patent-level inventive impact and a yearly top-tail impact indicator. The main executable panel used in the paper is rosenkopf_2001_analysis_dataset.csv and the principal coefficient export is rosenkopf_2001_model_coefficients.csv .

File	Role
rosenkopf_2001_analysis_dataset.csv	Patent-level search-type panel
rosenkopf_2001_model_coefficients.csv	Impact and top-5 coefficient table
rosenkopf_2001_sample_summary.csv	Search-type raw means

Core local datasets and result files used in the current paper.

3.1 Executable File Selection and Data Availability

The present manuscript works from the processed datasets and calculated exports already available in the workspace for Rosenkopf and Nerkar (2001). That does not mean the paper skips data analysis. It means the data analysis begins from the current executable panel, current coefficient export, and current sample-summary files rather than from a freshly rebuilt raw archive. The substantive obligation is therefore to explain exactly why each verified file is needed and how it enters the paper's argument. The analysis panel used in this paper is rosenkopf_2001_analysis_dataset.csv. Its role is

to estimate the local model and regenerate descriptive statistics. This file is not listed merely for bookkeeping. It is the concrete object that allows the manuscript to calculate the sample moments, estimators, and interpretation discussed in the main text. By naming the dataset rather than printing a filesystem path, the paper keeps the emphasis on evidence and function rather than on local machine details. The coefficient export used in this paper is rosenkopf_2001_model_coefficients.csv. Its role is to recover the reported local coefficients and p-values. This file is not listed merely for bookkeeping. It is the concrete object that allows the manuscript to calculate the sample moments, estimators, and interpretation discussed in the main text. By naming the dataset rather than printing a filesystem path, the paper keeps the emphasis on evidence and function rather than on local machine details. The sample summary used in this paper is rosenkopf_2001_sample_summary.csv. Its role is to audit sample construction and raw grouped moments. This file is not listed merely for bookkeeping. It is the concrete object that allows the manuscript to calculate the sample moments, estimators, and interpretation discussed in the main text. By naming the dataset rather than printing a filesystem path, the paper keeps the emphasis on evidence and function rather than on local machine details. Taken together, these files form the minimum executable evidence chain behind the paper. The theory comes from the source article, the estimable sample comes from the processed analysis panel, the regression narrative comes from the current coefficient export, and the sample audit comes from the local summary file. That is the basis on which the manuscript makes every empirical claim.

Stage	NeededFor	Dataset	Exists	Status
Source article	Recover theory, hypotheses, sample rules, definitions, tables, notes, and appendices	Strategic Management Journal - 2001 - Rosenkopf - Beyond local search boundary-spanning exploration and impact in the.pdf	No	Direct source
Analysis panel	Estimate the local model and regenerate descriptive statistics	rosenkopf_2001_analysis_dataset.csv	Yes	Executable local dataset
Coefficient export	Recover the reported local coefficients and p-values	rosenkopf_2001_model_coefficients.csv	Yes	Executable local result file
Sample summary	Audit sample construction and raw grouped moments	rosenkopf_2001_sample_summary.csv	Yes	Executable local result file

Verified current files used to support the paper's calculations.

3.2 Sample Reconstruction

The search-type file classifies patents into local, boundary-existing-field, and boundary-new-field search. Local patents account for 12,131 observations with mean impact 5.1357. Boundary-existing-field patents account for 3,270 observations with mean impact 4.5352. Boundary-new-field patents

account for 8,443 observations with mean impact 5.8602. Even before estimation, the raw means point to a substantive ranking in which the most novel form of search outperforms the more moderate boundary-spanning category. The top-tail shares follow the same direction: 0.0411 for local search, 0.0287 for boundary-existing-field search, and 0.0458 for boundary-new-field search.

SearchType	Patents	MeanImpact	Top5Share
boundary_new_field	8443.0000	5.8602	0.0458
local	12131.0000	5.1357	0.0411
boundary_existing_field	3270.0000	4.5352	0.0287

Sample reconstruction from the current local dataset.

3.3 Variable Construction and Derivation

The construct Local search is operationalized locally as Prior organizational experience and appears in the executable dataset as search_type = local. The derivation is not inferred loosely from the source article; it follows the verified local pipeline: Classify patents with positive prior organizational experience as local.. This mapping is retained because it preserves the construct's functional role in the original empirical argument while keeping the measurement auditable from the processed files used in estimation. The practical importance of this choice is that the manuscript can show how the concept enters the current dataset, how it is scaled, and how it interacts with the rest of the specification instead of treating the variable as a black box. In replication terms, this is the point where conceptual comparability and executable traceability have to be balanced carefully, and the chosen local definition is intended to preserve that balance rather than maximize superficial similarity. The construct Boundary-existing-field search is operationalized locally as No prior organizational but prior field experience and appears in the executable dataset as search_type = boundary_existing_field. The derivation is not inferred loosely from the source article; it follows the verified local pipeline: Classify patents with zero prior organizational but positive field experience as boundary-existing-field.. This mapping is retained because it preserves the construct's functional role in the original empirical argument while keeping the measurement auditable from the processed

files used in estimation. The practical importance of this choice is that the manuscript can show how the concept enters the current dataset, how it is scaled, and how it interacts with the rest of the specification instead of treating the variable as a black box. In replication terms, this is the point where conceptual comparability and executable traceability have to be balanced carefully, and the chosen local definition is intended to preserve that balance rather than maximize superficial similarity. The construct Boundary-new-field search is operationalized locally as No prior organizational and no prior field experience and appears in the executable dataset as search_type = boundary_new_field. The derivation is not inferred loosely from the source article; it follows the verified local pipeline: Classify patents with zero prior organizational and zero prior field experience as boundary-new-field.. This mapping is retained because it preserves the construct's functional role in the original empirical argument while keeping the measurement auditable from the processed files used in estimation. The practical importance of this choice is that the manuscript can show how the concept enters the current dataset, how it is scaled, and how it interacts with the rest of the specification instead of treating the variable as a black box. In replication terms, this is the point where conceptual comparability and executable traceability have to be balanced carefully, and the chosen local definition is intended to preserve that balance rather than maximize superficial similarity. The construct Inventive impact is operationalized locally as Patent-level citation impact and appears in the executable dataset as inventive_impact. The derivation is not inferred loosely from the source article; it follows the verified local pipeline: Use the local patent-level impact variable from the cleaned panel.. This mapping is retained because it preserves the construct's functional role in the original empirical argument while keeping the measurement auditable from the processed files used in estimation. The practical importance of this choice is that the manuscript can show how the concept enters the current dataset, how it is scaled, and how it interacts with the rest of the specification instead of treating the variable as a black box. In replication terms, this is the point where conceptual comparability and executable traceability have to be balanced carefully, and the chosen local definition is intended to preserve that balance rather than maximize superficial similarity. The construct Tail impact is operationalized locally as Top-5-percent yearly impact indicator and appears in the executable dataset as high_top5_y. The derivation is not inferred loosely from the source article; it follows the verified local pipeline: Use the already materialized yearly top-5-percent indicator.. This mapping is retained because it preserves the construct's func-

tional role in the original empirical argument while keeping the measurement auditable from the processed files used in estimation. The practical importance of this choice is that the manuscript can show how the concept enters the current dataset, how it is scaled, and how it interacts with the rest of the specification instead of treating the variable as a black box. In replication terms, this is the point where conceptual comparability and executable traceability have to be balanced carefully, and the chosen local definition is intended to preserve that balance rather than maximize superficial similarity.

OriginalConstruct	LocalMeasure	ExactVariable	Derivation
Local search	Prior organizational experience	search_type = local	Classify patents with positive prior organizational experience as local.
Boundary-existing-field search	No prior organizational but prior field experience	search_type = boundary_existing_field	Classify patents with zero prior organizational but positive field experience as boundary-existing-field.
Boundary-new-field search	No prior organizational and no prior field experience	search_type = boundary_new_field	Classify patents with zero prior organizational and zero prior field experience as boundary-new-field.

OriginalConstruct	LocalMeasure	ExactVariable	Derivation
Inventive impact	Patent-level citation impact	inventive_impact	Use the local patent-level impact variable from the cleaned panel.
Tail impact	Top-5-percent yearly impact indicator	high_top5_y	Use the already materialized yearly top-5-percent indicator.

Variable mapping and derivation logic in the main text.

3.4 Construct-Level Derivation Logic

The derivation appendix records how each major construct enters the executable estimand for Rosenkopf and Nerkar (2001). The point is not to rehearse notation for its own sake, but to show how the source paper's conceptual vocabulary is translated into verified columns in the local panel. For Local search, the manuscript uses Prior organizational experience under the exact local implementation search_type = local. The derivation step is: Classify patents with positive prior organizational experience as local.. This construct is retained because it occupies the same analytic position in the local design that the published paper assigns to the original concept. For Boundary-existing-field search, the manuscript uses No prior organizational but prior field experience under the exact local implementation search_type = boundary_existing_field. The derivation step is: Classify patents with zero prior organizational but positive field experience as boundary-existing-field.. This construct is retained because it occupies the same analytic position in the local design that the published paper assigns to the original concept. For Boundary-new-field search, the manuscript uses No prior organizational and no prior field experience under the exact local

implementation search_type = boundary_new_field. The derivation step is: Classify patents with zero prior organizational and zero prior field experience as boundary-new-field.. This construct is retained because it occupies the same analytic position in the local design that the published paper assigns to the original concept. For Inventive impact, the manuscript uses Patent-level citation impact under the exact local implementation inventive_impact. The derivation step is: Use the local patent-level impact variable from the cleaned panel.. This construct is retained because it occupies the same analytic position in the local design that the published paper assigns to the original concept. For Tail impact, the manuscript uses Top-5-percent yearly impact indicator under the exact local implementation high_top5_y. The derivation step is: Use the already materialized yearly top-5-percent indicator.. This construct is retained because it occupies the same analytic position in the local design that the published paper assigns to the original concept.

4 Descriptive Analysis of the Current Dataset

The current dataset yields a richer descriptive picture than a single mean or coefficient can convey. Inventive impact has mean 5.3099, standard deviation 17.5532, and an interquartile range from 1.0000 to 6.0000; Top-5 indicator has mean 0.0411, standard deviation 0.1985, and an interquartile range from 0.0000 to 0.0000; Breadth of recombination has mean 4.2941, standard deviation 4.5669, and an interquartile range from 2.0000 to 5.0000; Recombination degree has mean 0.5549, standard deviation 0.1738, and an interquartile range from 0.4444 to 0.6914 . The time-series view is equally informative. V1 changes from 643.0000 in the first observed period to 534.0000 in the last observed period; V2 changes from 4.0202 in the first observed period to 3.7678 in the last observed period; V3 changes from 2.8694 in the first observed period to 4.6292 in the last observed period . These temporal shifts help explain why a paper-comparable replication has to describe the support of the data before turning to regression output. The grouped search taxonomy reveals the raw ranking that motivates the later models. Boundary-new-field search has mean impact 5.8602 and top-tail share 0.0458, local search has mean impact 5.1357 and top-tail share 0.0411, and boundary-existing-field search has mean impact 4.5352 and top-tail share 0.0287. The correlation structure is reported in a compact form because it helps distinguish whether the main explanatory variables move in

lockstep with the controls or whether the regression is drawing on independent variation. That is especially important in these replications, where construct translation can otherwise make the specification look more deterministic than it really is.

4.1 Data Quality, Coverage, and Support

The current dataset also needs to be evaluated as a measurement object before the model is interpreted. Inventive impact has missing share 0.00; Top-5 indicator has missing share 0.00; Breadth of recombination has missing share 0.00 . At the other end of the support, Inventive impact remains close to complete with missing share 0.00; Top-5 indicator remains close to complete with missing share 0.00; Breadth of recombination remains close to complete with missing share 0.00 . This matters because the credibility of the regression depends on whether the focal constructs survive sample cleaning with enough support to be interpreted substantively. Coverage across entities is also visible in the current panel rather than assumed. 6066 contributes 3,200 observations; 5047 contributes 1,318 observations; 4194 contributes 1,017 observations; 7435 contributes 854 observations; 4087 contributes 830 observations . The purpose of reporting these concentrations is to show whether the estimated mechanism is being learned from a broad panel or from a narrow set of heavily observed entities. The support of the focal regressor is broad enough to permit a distributional reading in addition to the formal model. In the lowest observed quantile, the mean regressor is 2.4112 and the mean outcome is 4.4643 across 13,675 observations. In the highest observed quantile, the corresponding values are 9.9635, 8.1151, and 4,438 observations. This shift matters because it shows that the regression is being estimated on economically different parts of the observed support rather than on nearly identical cells. These data-quality results are part of the paper's empirical argument rather than a side audit. The replication matters because boundary-spanning search claims are often treated broadly, while the original paper distinguishes among qualitatively different forms of exploratory search. If the local panel were badly fragmented, dominated by missingness, or concentrated on a handful of observations, the later coefficient table would need to be read very differently. Bringing this evidence into the body of the article is therefore necessary for an honest replication.

Variable	Missing	MissingShare
Inventive impact	0.0000	0.0000
Top-5 indicator	0.0000	0.0000
Breadth of recombination	0.0000	0.0000
Recombination degree	0.0000	0.0000
Field experience	0.0000	0.0000
Organizational experience	0.0000	0.0000
Prior-art citations	0.0000	0.0000
Same-year patents	0.0000	0.0000
Prior cumulative stock	0.0000	0.0000

Missingness profile for variables analyzed in the body of the paper.

Variable	N	Mean	SD	P25	Median	P75	Min	M
Inventive impact	23844.0000	5.3099	17.5532	1.0000	2.0000	6.0000	0.0000	812.00
Top-5 indicator	23844.0000	0.0411	0.1985	0.0000	0.0000	0.0000	0.0000	1.00
Breadth of recombination	23844.0000	4.2941	4.5669	2.0000	3.0000	5.0000	2.0000	117.00
Recombination degree	23844.0000	0.5549	0.1738	0.4444	0.5600	0.6914	0.0454	0.95
Field experience	23844.0000	1.6339	1.8203	0.0000	1.0986	2.7726	0.0000	8.01
Organizational experience	23844.0000	0.9227	1.3174	0.0000	0.6931	1.3863	0.0000	7.65
Prior-art citations	23844.0000	15.0909	50.4465	5.0000	8.0000	13.0000	2.0000	2263.00
Same-year patents	23844.0000	728.7229	1302.8291	109.0000	313.0000	652.0000	1.0000	7321.00

Variable	N	Mean	SD	P25	Median	P75	Min	M
Prior cumulative stock	23844.0000	7.5074	2.2142	6.3297	7.6930	8.8934	0.0000	11.32

Main descriptive statistics discussed in the body of the paper.

Period	V1	V2	V3	V4	V5
1986.0000	643.0000	4.0202	2.8694	0.5175	0.4277
1987.0000	789.0000	4.2801	2.9176	0.5177	0.4335
1988.0000	1066.0000	4.4606	3.1323	0.5319	0.4353
1989.0000	1390.0000	3.9338	3.0813	0.5272	0.4007
1990.0000	1349.0000	4.5137	3.1156	0.5218	0.3744
1991.0000	1325.0000	4.7381	3.1502	0.5314	0.3743
1992.0000	1302.0000	5.0154	3.3318	0.5346	0.4040
1993.0000	1354.0000	5.6721	3.4609	0.5323	0.3877
1994.0000	1330.0000	6.2654	3.5782	0.5413	0.3932
1995.0000	1217.0000	6.2597	3.8439	0.5563	0.3780
1996.0000	659.0000	6.7269	3.8376	0.5306	0.3763
1997.0000	646.0000	6.0139	4.2446	0.5521	0.4195
1998.0000	356.0000	6.2949	3.9242	0.5473	0.4045
1999.0000	654.0000	5.4832	3.8104	0.5436	0.3700
2000.0000	636.0000	5.5409	4.0220	0.5402	0.3632
2001.0000	614.0000	5.1515	4.4137	0.5625	0.4169
2002.0000	632.0000	4.8892	4.3845	0.5578	0.3687
2003.0000	522.0000	4.5019	4.4176	0.5505	0.3487
2004.0000	324.0000	3.6636	4.8086	0.5734	0.4136
2005.0000	284.0000	4.5000	5.4859	0.5763	0.4085
2006.0000	336.0000	4.8899	5.2649	0.5661	0.3214
2007.0000	246.0000	3.6545	5.2683	0.5692	0.3089

Period	V1	V2	V3	V4	V5
2008.0000	535.0000	3.6523	4.5121	0.5904	0.3607
2009.0000	206.0000	4.8010	6.0049	0.5992	0.2961
2010.0000	511.0000	4.4168	6.6477	0.6078	0.2564
2011.0000	651.0000	6.4516	6.2750	0.6090	0.2550
2012.0000	1068.0000	4.0000	6.5187	0.6113	0.2603
2013.0000	868.0000	5.3859	6.7304	0.5978	0.2604
2014.0000	725.0000	12.6428	6.9890	0.6000	0.2290
2015.0000	1072.0000	6.6157	6.1744	0.5971	0.2043
2016.0000	534.0000	3.7678	4.6292	0.5811	0.1667

Observed time patterns in the current dataset.

SearchType	Patents	MeanImpact	Top5Share
boundary_new_field	8443.0000	5.8602	0.0458
local	12131.0000	5.1357	0.0411
boundary_existing_field	3270.0000	4.5352	0.0287

Grouped summaries used to describe raw patterns in the current dataset.

Variable	Inventive impact	Top-5 indicator	Breadth of recombination	Recombination degree	Field experience	Organizational experience	Prior-art citations	Same-year patents	Prior cumulative stock
Inventive impact	1.000	0.455	0.364	0.018	-0.021	-0.008	0.591	-0.018	0.022
Top-5 indicator	0.455	1.000	0.129	0.024	-0.012	0.006	0.149	-0.015	0.005
Breadth of recombination	0.364	0.129	1.000	0.466	-0.263	-0.144	0.822	0.011	0.149

Variable	Inventive impact	Top-5 indicator	Breadth of recombination	Recombination degree	Field experience	Organizational experience	Prior-art citations	Same-year patents	Prior cumulative stock
Recombination degree	0.018	0.024	0.466	1.000	-0.456	-0.313	0.140	0.065	0.122
Field experience	-0.021	-0.012	-0.263	-0.456	1.000	0.838	-0.077	0.060	0.132
Organizational experience	-0.008	0.006	-0.144	-0.313	0.838	1.000	-0.018	0.152	0.267

Compact correlation structure for the main variables.

Entity	Observations	FirstPeriod	LastPeriod
6066.0000	3200.0000	1986.0000	2016.0000
5047.0000	1318.0000	1986.0000	2016.0000
4194.0000	1017.0000	1986.0000	2010.0000
7435.0000	854.0000	1986.0000	2016.0000
4087.0000	830.0000	1986.0000	2016.0000
4839.0000	815.0000	1986.0000	2016.0000
7585.0000	774.0000	1986.0000	2016.0000
10499.0000	755.0000	1986.0000	2016.0000
5073.0000	671.0000	1986.0000	2008.0000
4060.0000	574.0000	1986.0000	2016.0000
1300.0000	551.0000	1986.0000	2016.0000
8762.0000	524.0000	1986.0000	2016.0000
1690.0000	452.0000	1990.0000	2016.0000
1581.0000	421.0000	1986.0000	2004.0000
10983.0000	403.0000	1986.0000	2016.0000
12141.0000	398.0000	1995.0000	2016.0000

Entity	Observations	FirstPeriod	LastPeriod
4503.0000	344.0000	1986.0000	2016.0000
6266.0000	340.0000	1986.0000	2016.0000
2285.0000	334.0000	1986.0000	2016.0000
7475.0000	300.0000	1986.0000	1998.0000

Most observed entities in the current dataset.

5 Empirical Strategy and Identification

The estimand targeted in this manuscript is Patent-level impact and top-tail models with search-type indicators and patent-level controls. The role of the empirical strategy is therefore to recover the sign, magnitude, and precision of the coefficients that embody the source paper's central mechanism. Identification remains observational. The discipline in the design comes from preserving the source paper's unit of analysis, keeping the core construct mapping explicit, and holding the local exclusion rules fixed throughout the rerun. This does not convert the study into an experimental design, but it does make the inferential claim narrower and more honest. The first set of replication decisions concerns scope. The unit of analysis is Patent, the sample frame is Patent-level recombination panel derived from the cleaned local reproduction dataset., and the empirical window is 1986 to 2016 in the local executable panel.. These choices define what counts as a comparable observation in the local reconstruction and prevent the paper from drifting into a broader but theoretically mismatched sample. The second set of decisions concerns measurement and specification. The outcome is Patent-level inventive impact and a yearly top-tail impact indicator., and the main explanatory design is Patent-level impact models with search-type indicators comparing boundary-spanning forms to local search.. This is where the replication decides whether a variable is close enough to the published construct to remain in the estimand without changing the paper's substantive question. The third set of decisions concerns statistical discipline. Controls and fixed effects are handled as follows: Patent-level controls for breadth, degree, citations, scale, and prior stock with available fixed effects.. The crucial exclusion or substitution rule is: The original

industry-specific coding is not locally recoverable, so the executable rerun generalizes the taxonomy using prior organizational and field experience counts.. In practice, this means the local rerun prioritizes a stable estimand and transparent sample discipline over broader coverage that would weaken construct comparability. A final design choice concerns interpretation. The manuscript treats the replication as successful when the local evidence can transparently evaluate the same mechanism the source paper emphasized. When the local data architecture differs, the paper marks the result as a calculated analogue rather than claiming byte-identical duplication.

DecisionPoint	LocalSpecification
Research question	The source paper asks whether boundary-spanning search improves inventive impact relative to local search.
Unit of analysis	Patent
Sample frame	Patent-level recombination panel derived from the cleaned local reproduction dataset.
Time window	1986 to 2016 in the local executable panel.
Outcome definition	Patent-level inventive impact and a yearly top-tail impact indicator.
Main explanatory design	Patent-level impact models with search-type indicators comparing boundary-spanning forms to local search.

DecisionPoint	LocalSpecification
Controls and fixed effects	Patent-level controls for breadth, degree, citations, scale, and prior stock with available fixed effects.
Exclusion or substitution logic	The original industry-specific coding is not locally recoverable, so the executable rerun generalizes the taxonomy using prior organizational and field experience counts.

Replication decision register retained in the main text.

5.1 Why These Design Decisions Are Necessary

Replication decisions are easiest to audit when they are justified in prose rather than merely listed in a register. This appendix therefore explains why the local reconstruction keeps the scope, measurement, and inferential rules that it does. Scope decisions: the manuscript works at the level of Patent inside the sample frame Patent-level recombination panel derived from the cleaned local reproduction dataset. over 1986 to 2016 in the local executable panel.. These are not cosmetic constraints. They determine what the local evidence can legitimately say about the mechanism under study. Measurement decisions: the outcome is Patent-level inventive impact and a yearly top-tail impact indicator. and the main explanatory design is Patent-level impact models with search-type indicators comparing boundary-spanning forms to local search.. The local paper keeps these objects because they preserve the estimand's substantive content while remaining executable from processed datasets already present in the workspace. Inference decisions: Patent-level controls for breadth, degree, citations, scale, and prior stock with available fixed effects. The main exclusion or

substitution rule is The original industry-specific coding is not locally recoverable, so the executable rerun generalizes the taxonomy using prior organizational and field experience counts. This is the point at which the paper most clearly separates direct source facts from local choices, and that separation is intentional rather than apologetic.

$$E[\text{Impact}_{pft} \mid X_{pft}] = \exp\left(\beta_1 \mathbb{1}[\text{BoundaryExisting}_p] + \beta_2 \mathbb{1}[\text{BoundaryNew}_p] + \beta_3 X_{pft} + \alpha_f + \gamma_t\right). \tag{1}$$

6 Results

6.1 Main Estimates

The central local estimate is -0.0389 on boundary_existing_field vs local (p = 0.3868). The local analogue supports the most novel boundary-spanning channel much more clearly than the moderate boundary-spanning channel. Boundary new field versus local is estimated at 0.1126 with p = 0.0014. Recombination degree is estimated at -0.4108 with p = 0.0226. Prior-art citations is estimated at 0.0013 with p = 0.0548. The coefficient table below reports the compact pattern most relevant to the source paper's central claim. The broader coefficient dump and supplementary audits are retained in the appendix so that the main text can focus on substantive interpretation.

model	Term	estimate	std_error	p_value
rosenkopf_impact	Boundary existing field versus local	-0.0389	0.0449	0.3868
rosenkopf_impact	Boundary new field versus local	0.1126	0.0353	0.0014
rosenkopf_impact	Breadth of recombination	0.0223	0.0130	0.0870
rosenkopf_impact	Recombination degree	-0.4108	0.1802	0.0226
rosenkopf_impact	Prior-art citations	0.0013	0.0007	0.0548

model	Term	estimate	std_error	p_value
rosenkopf_impact	Log prior patent stock	-0.1965	0.1067	0.0657

Key coefficient table used in the main text.

6.2 Detailed Interpretation of the Local Results

The coefficient pattern is not being summarized as a bare sign check. Boundary new field versus local is estimated at 0.1126 with a p-value of 0.0014; Recombination degree is estimated at -0.4108 with a p-value of 0.0226; Prior-art citations is estimated at 0.0013 with a p-value of 0.0548; Log prior patent stock is estimated at -0.1965 with a p-value of 0.0657 . Within the focal specification, the sharpest local signal comes from Boundary new field versus local, which is estimated at 0.1126 with p = 0.0014. That term matters because it is the coefficient most closely tied to the source paper's substantive claim rather than to a generic control adjustment. The scale of the underlying variables also matters for interpretation. The outcome Inventive impact has mean 5.3099 and standard deviation 17.5532 in the local panel, while the main regressor Breadth of recombination has mean 4.2941 and standard deviation 4.5669. A simple grouped comparison also helps clarify the pattern. Across the regressor quantiles, the mean outcome ranges from 4.4643 to 8.1151 while the mean regressor ranges from 2.4112 to 9.9635. This does not substitute for the model, but it shows how much raw variation the specification must explain. Taken together, the descriptive evidence and formal estimates should be read as a structured assessment of the paper's mechanism. The local analogue supports the most novel boundary-spanning channel much more clearly than the moderate boundary-spanning channel. The substantive question is not whether every coefficient matches a published table entry digit for digit, but whether the local evidence supports the same directional and conceptual interpretation.

6.3 Coefficient Families in the Current Estimation Output

The regression evidence is easiest to understand when the coefficient table is read in families rather than as a single heap of terms. The paper therefore separates the estimates by model and highlights the strongest positive, strongest negative, and most focal signals within each specification. Within rosenkopf_impact, the leading coefficient family reads as follows: Boundary new field versus local = 0.1126 (p = 0.0014); Recombination degree = -0.4108 (p = 0.0226); total_patents_same_year = -0.0001 (p = 0.0358); Prior-art citations = 0.0013 (p = 0.0548); Log prior patent stock = -0.1965 (p = 0.0657). The largest positive estimate in this specification is Boundary new field versus local at 0.1126, while the largest negative estimate is Recombination degree at -0.4108. Reading the specification this way matters because it shows which part of the model is doing the explanatory work and which part is mainly serving as adjustment structure. Within rosenkopf_top5, the leading coefficient family reads as follows: Log prior patent stock = -0.4208 (p = 0.0184); Boundary existing field versus local = -0.2317 (p = 0.1534); Prior-art citations = 0.0055 (p = 0.1746); total_patents_same_year = -0.0001 (p = 0.3360); Boundary new field versus local = 0.1039 (p = 0.3585). The largest positive estimate in this specification is Boundary new field versus local at 0.1039, while the largest negative estimate is Log prior patent stock at -0.4208. Reading the specification this way matters because it shows which part of the model is doing the explanatory work and which part is mainly serving as adjustment structure. This fuller coefficient reading is especially important for Rosenkopf and Nerkar (2001) because the source paper's contribution is not exhausted by one headline estimate. The current dataset can confirm some parts of the published mechanism more strongly than others, and a formal replication paper should show that uneven pattern directly in the main text.

6.4 Model Comparison Using the Current Dataset

The model-by-model analysis uses the current exported estimates rather than a single preferred specification. In rosenkopf_impact, the most informative focal term is Boundary new field versus local with estimate 0.1126 and p = 0.0014; In rosenkopf_top5, the most informative focal term is Log prior patent stock with estimate -0.4208 and p = 0.0184 . The rosenkopf_impact specification

contributes its own reading to the replication. Boundary new field versus local is estimated at 0.1126 with p = 0.0014 and is classified as a strong signal; Recombination degree is estimated at -0.4108 with p = 0.0226 and is classified as a moderate signal; Prior-art citations is estimated at 0.0013 with p = 0.0548 and is classified as a suggestive signal . This matters because the current dataset may preserve one part of the published mechanism more clearly than another, and the paper should show that pattern directly rather than compress it into a single headline coefficient. The rosenkopf_top5 specification contributes its own reading to the replication. Log prior patent stock is estimated at -0.4208 with p = 0.0184 and is classified as a moderate signal; Boundary existing field versus local is estimated at -0.2317 with p = 0.1534 and is classified as a weak signal; Prior-art citations is estimated at 0.0055 with p = 0.1746 and is classified as a weak signal . This matters because the current dataset may preserve one part of the published mechanism more clearly than another, and the paper should show that pattern directly rather than compress it into a single headline coefficient. This model comparison matters for Rosenkopf and Nerkar (2001) because the source article's claim can only be evaluated responsibly if the pattern persists across the outcome or specification variants that are actually present in the local results folder. Put differently, the current dataset should be allowed to show where the mechanism is stable, where it weakens, and where it changes sign under a different outcome definition.

Term	Model	Estimate	PValue	Signal	Reading
Boundary existing field versus local	rosenkopf_impact	-0.0389	0.3868	Weak	Negative estimate with p = 0.3868.
Boundary new field versus local	rosenkopf_impact	0.1126	0.0014	Strong	Positive estimate with p = 0.0014.

Term	Model	Estimate	PValue	Signal	Reading
Breadth of recombination	rosenkopf_impact	0.0223	0.0870	Suggestive	Positive estimate with p = 0.0870.
Recombination degree	rosenkopf_impact	-0.4108	0.0226	Moderate	Negative estimate with p = 0.0226.
Prior-art citations	rosenkopf_impact	0.0013	0.0548	Suggestive	Positive estimate with p = 0.0548.
Log prior patent stock	rosenkopf_impact	-0.1965	0.0657	Suggestive	Negative estimate with p = 0.0657.
Boundary existing field versus local	rosenkopf_top5	-0.2317	0.1534	Weak	Negative estimate with p = 0.1534.
Boundary new field versus local	rosenkopf_top5	0.1039	0.3585	Weak	Positive estimate with p = 0.3585.
Breadth of recombination	rosenkopf_top5	0.0069	0.7349	Weak	Positive estimate with p = 0.7349.

Term	Model	Estimate	PValue	Signal	Reading
Recombination degree	rosenkopf_top5	-0.0597	0.8303	Weak	Negative estimate with p = 0.8303.
Prior-art citations	rosenkopf_top5	0.0055	0.1746	Weak	Positive estimate with p = 0.1746.
Log prior patent stock	rosenkopf_top5	-0.4208	0.0184	Moderate	Negative estimate with p = 0.0184.

Model-by-model interpretation table drawn from the current coefficient exports.

6.5 Diagnostic Evidence

A separate model-fit export is not available for this paper, so the interpretation relies on the coefficient file and the observed distribution of the current dataset.

term	estimate	std_error	statistic	p_value	model
search_type::boundary_existing_field	-0.0389	0.0449	-0.8655	0.3868	rosenkopf_impact
search_type::boundary_new_field	0.1126	0.0353	3.1882	0.0014	rosenkopf_impact
breadth_recombination	0.0223	0.0130	1.7114	0.0870	rosenkopf_impact
recombination_degree	-0.4108	0.1802	-2.2800	0.0226	rosenkopf_impact
n_prior_art_cites	0.0013	0.0007	1.9206	0.0548	rosenkopf_impact
total_patents_same_year	-0.0001	0.0000	-2.0994	0.0358	rosenkopf_impact
total_patents_until_prev_log	-0.1965	0.1067	-1.8408	0.0657	rosenkopf_impact

term	estimate	std_error	statistic	p_value	model
search_type::boundary_existing_field	-0.2317	0.1623	-1.4276	0.1534	rosenkopf_top5
search_type::boundary_new_field	0.1039	0.1131	0.9183	0.3585	rosenkopf_top5
breadth_recombination	0.0069	0.0205	0.3386	0.7349	rosenkopf_top5
recombination_degree	-0.0597	0.2786	-0.2143	0.8303	rosenkopf_top5
n_prior_art_cites	0.0055	0.0041	1.3577	0.1746	rosenkopf_top5
total_patents_same_year	-0.0001	0.0001	-0.9622	0.3360	rosenkopf_top5
total_patents_until_prev_yr	-0.4208	0.1786	-2.3566	0.0184	rosenkopf_top5

Full coefficient table from the current local estimation output.

7 Comparison to the Published Study

The comparison to the published paper is intentionally narrow. The local evidence is strongest for boundary-new-field search rather than for boundary-existing-field search. Boundary-new-field patents receive higher average impact relative to local search, while boundary-existing-field patents are lower and imprecise. The manuscript therefore evaluates concordance at the level of mechanism, sign pattern, and construct behavior before drawing any conclusion about exact numerical replication. This comparison rule matters because the local paper is based only on processed datasets and verified outputs already present in the workspace. When the original paper depends on proprietary, historical, or otherwise unavailable preprocessing layers, a full byte-identical recreation is not claimed. The more defensible question is whether the local evidence would lead a careful reader to a similar substantive interpretation.

8 Discussion

The local evidence is strongest for boundary-new-field search rather than for boundary-existing-field search. Boundary-new-field patents receive higher average impact relative to local search,

while boundary-existing-field patents are lower and imprecise. Taken together, the results are most informative about whether the source paper's mechanism is directionally recoverable in local data rather than whether every original coefficient can be duplicated exactly. The first limitation concerns measurement. The local search-type coding generalizes the original industry-specific implementation. The second limitation concerns sample translation. Mismatch with published estimates can arise from both sample expansion and measurement differences. The third limitation concerns interpretation. The exercise is most informative about directional mechanisms.

9 Conclusion

The paper's contribution is a transparent three-way search taxonomy on a local executable panel that separates local, boundary-existing-field, and boundary-new-field search. The paper therefore contributes a traceable local replication record that can be rerun, audited, and extended as additional datasets or original-source details become available.

A Appendix A: Supplementary Sample Reconstruction

The sample table below shows the observable route from the broader local file universe to the final estimating sample. It is included to make sample narrowing visible and to show where exclusions enter the paper.

SearchType	Patents	MeanImpact	Top5Share
boundary_new_field	8443.0000	5.8602	0.0458
local	12131.0000	5.1357	0.0411
boundary_existing_field	3270.0000	4.5352	0.0287

Detailed sample reconstruction table.

B Appendix B: Supplementary Descriptive Tables

The descriptive appendix documents the scale, spread, and support of the variables used in the paper. These moments matter because they determine how the estimated coefficients should be read in substantive rather than merely statistical terms.

Variable	N	Mean	SD	P25	Median	P75	Min	M
Inventive impact	23844.0000	5.3099	17.5532	1.0000	2.0000	6.0000	0.0000	812.00
Top-5 indicator	23844.0000	0.0411	0.1985	0.0000	0.0000	0.0000	0.0000	1.00
Breadth of recombination	23844.0000	4.2941	4.5669	2.0000	3.0000	5.0000	2.0000	117.00
Recombination degree	23844.0000	0.5549	0.1738	0.4444	0.5600	0.6914	0.0454	0.95
Field experience	23844.0000	1.6339	1.8203	0.0000	1.0986	2.7726	0.0000	8.01
Organizational experience	23844.0000	0.9227	1.3174	0.0000	0.6931	1.3863	0.0000	7.65
Prior-art citations	23844.0000	15.0909	50.4465	5.0000	8.0000	13.0000	2.0000	2263.00
Same-year patents	23844.0000	728.7229	1302.8291	109.0000	313.0000	652.0000	1.0000	7321.00
Prior cumulative stock	23844.0000	7.5074	2.2142	6.3297	7.6930	8.8934	0.0000	11.32

Descriptive statistics appendix.

C Appendix C: Correlation Matrix

Variable	Inventive impact	Top-5 indicator	Breadth of recombination	Recombination degree	Field experience	Organizational experience	Prior-art citations	Same-year patents	Prior cumulative stock
Inventive impact	1.000	0.455	0.364	0.018	-0.021	-0.008	0.591	-0.018	0.022
Top-5 indicator	0.455	1.000	0.129	0.024	-0.012	0.006	0.149	-0.015	0.005
Breadth of recombination	0.364	0.129	1.000	0.466	-0.263	-0.144	0.822	0.011	0.149
Recombination degree	0.018	0.024	0.466	1.000	-0.456	-0.313	0.140	0.065	0.122
Field experience	-0.021	-0.012	-0.263	-0.456	1.000	0.838	-0.077	0.060	0.132
Organizational experience	-0.008	0.006	-0.144	-0.313	0.838	1.000	-0.018	0.152	0.267
Prior-art citations	0.591	0.149	0.822	0.140	-0.077	-0.018	1.000	-0.016	0.081
Same-year patents	-0.018	-0.015	0.011	0.065	0.060	0.152	-0.016	1.000	0.520
Prior cumulative stock	0.022	0.005	0.149	0.122	0.132	0.267	0.081	0.520	1.000

Correlation matrix for the main numeric variables.

D Appendix D: Yearly Audit

The yearly audit provides a simple check on whether the local panel is driven by a narrow time slice or whether the variables and sample counts evolve over a broad observed window.

Period	V1	V2	V3	V4	V5
1986.0000	643.0000	4.0202	2.8694	0.5175	0.4277
1987.0000	789.0000	4.2801	2.9176	0.5177	0.4335
1988.0000	1066.0000	4.4606	3.1323	0.5319	0.4353
1989.0000	1390.0000	3.9338	3.0813	0.5272	0.4007
1990.0000	1349.0000	4.5137	3.1156	0.5218	0.3744
1991.0000	1325.0000	4.7381	3.1502	0.5314	0.3743
1992.0000	1302.0000	5.0154	3.3318	0.5346	0.4040
1993.0000	1354.0000	5.6721	3.4609	0.5323	0.3877
1994.0000	1330.0000	6.2654	3.5782	0.5413	0.3932
1995.0000	1217.0000	6.2597	3.8439	0.5563	0.3780
1996.0000	659.0000	6.7269	3.8376	0.5306	0.3763
1997.0000	646.0000	6.0139	4.2446	0.5521	0.4195
1998.0000	356.0000	6.2949	3.9242	0.5473	0.4045
1999.0000	654.0000	5.4832	3.8104	0.5436	0.3700
2000.0000	636.0000	5.5409	4.0220	0.5402	0.3632
2001.0000	614.0000	5.1515	4.4137	0.5625	0.4169
2002.0000	632.0000	4.8892	4.3845	0.5578	0.3687
2003.0000	522.0000	4.5019	4.4176	0.5505	0.3487
2004.0000	324.0000	3.6636	4.8086	0.5734	0.4136
2005.0000	284.0000	4.5000	5.4859	0.5763	0.4085
2006.0000	336.0000	4.8899	5.2649	0.5661	0.3214
2007.0000	246.0000	3.6545	5.2683	0.5692	0.3089
2008.0000	535.0000	3.6523	4.5121	0.5904	0.3607
2009.0000	206.0000	4.8010	6.0049	0.5992	0.2961

Period	V1	V2	V3	V4	V5
2010.0000	511.0000	4.4168	6.6477	0.6078	0.2564
2011.0000	651.0000	6.4516	6.2750	0.6090	0.2550
2012.0000	1068.0000	4.0000	6.5187	0.6113	0.2603
2013.0000	868.0000	5.3859	6.7304	0.5978	0.2604
2014.0000	725.0000	12.6428	6.9890	0.6000	0.2290
2015.0000	1072.0000	6.6157	6.1744	0.5971	0.2043
2016.0000	534.0000	3.7678	4.6292	0.5811	0.1667

Year-by-year sample audit.

E Appendix E: Grouped Outcome Patterns

Grouped summaries do not replace the formal model, but they show the raw outcome structure that the specification is attempting to explain. For that reason they are retained here as descriptive support rather than promoted to the status of a main estimate.

Quantile	N	MeanRegressor	MeanOutcome
1.0000	13675.0000	2.4112	4.4643
2.0000	3458.0000	4.0000	4.9251
3.0000	2273.0000	5.0000	5.5055
4.0000	4438.0000	9.9635	8.1151

Quantile summary for the main regressor.

SearchType	Patents	MeanImpact	Top5Share
boundary_new_field	8443.0000	5.8602	0.0458
local	12131.0000	5.1357	0.0411
boundary_existing_field	3270.0000	4.5352	0.0287

Raw group means from the local summary file.

F Appendix F: Full Coefficient Table and Fit Statistics

The full coefficient export is included for completeness so that readers can see the focal terms in the context of the broader specification. Where a fit file is available, it is appended here rather than in the main text to keep the article narrative readable.

term	estimate	std_error	statistic	p_value	model
search_type::boundary_existing_field	-0.0389	0.0449	-0.8655	0.3868	rosenkopf_impact
search_type::boundary_new_field	0.1261	0.0353	3.1882	0.0014	rosenkopf_impact
breadth_recombination	0.0223	0.0130	1.7114	0.0870	rosenkopf_impact
recombination_degree	-0.4108	0.1802	-2.2800	0.0226	rosenkopf_impact
n_prior_art_cites	0.0013	0.0007	1.9206	0.0548	rosenkopf_impact
total_patents_same_year	-0.0001	0.0000	-2.0994	0.0358	rosenkopf_impact
total_patents_until_prev_log	-0.1965	0.1067	-1.8408	0.0657	rosenkopf_impact
search_type::boundary_existing_field	-0.2317	0.1623	-1.4276	0.1534	rosenkopf_top5
search_type::boundary_new_field	0.1039	0.1131	0.9183	0.3585	rosenkopf_top5
breadth_recombination	0.0069	0.0205	0.3386	0.7349	rosenkopf_top5
recombination_degree	-0.0597	0.2786	-0.2143	0.8303	rosenkopf_top5
n_prior_art_cites	0.0055	0.0041	1.3577	0.1746	rosenkopf_top5
total_patents_same_year	-0.0001	0.0001	-0.9622	0.3360	rosenkopf_top5
total_patents_until_prev_log	-0.4208	0.1786	-2.3566	0.0184	rosenkopf_top5

Full coefficient dump exported from the local estimation script.

G Appendix G: Coverage Tables

Variable	Missing	MissingShare
Inventive impact	0.0000	0.0000
Top-5 indicator	0.0000	0.0000
Breadth of recombination	0.0000	0.0000
Recombination degree	0.0000	0.0000
Field experience	0.0000	0.0000
Organizational experience	0.0000	0.0000
Prior-art citations	0.0000	0.0000
Same-year patents	0.0000	0.0000
Prior cumulative stock	0.0000	0.0000

Missingness audit for the variables summarized in the manuscript.

Entity	Observations	FirstPeriod	LastPeriod
6066.0000	3200.0000	1986.0000	2016.0000
5047.0000	1318.0000	1986.0000	2016.0000
4194.0000	1017.0000	1986.0000	2010.0000
7435.0000	854.0000	1986.0000	2016.0000
4087.0000	830.0000	1986.0000	2016.0000
4839.0000	815.0000	1986.0000	2016.0000
7585.0000	774.0000	1986.0000	2016.0000
10499.0000	755.0000	1986.0000	2016.0000
5073.0000	671.0000	1986.0000	2008.0000
4060.0000	574.0000	1986.0000	2016.0000
1300.0000	551.0000	1986.0000	2016.0000
8762.0000	524.0000	1986.0000	2016.0000
1690.0000	452.0000	1990.0000	2016.0000

Entity	Observations	FirstPeriod	LastPeriod
1581.0000	421.0000	1986.0000	2004.0000
10983.0000	403.0000	1986.0000	2016.0000
12141.0000	398.0000	1995.0000	2016.0000
4503.0000	344.0000	1986.0000	2016.0000
6266.0000	340.0000	1986.0000	2016.0000
2285.0000	334.0000	1986.0000	2016.0000
7475.0000	300.0000	1986.0000	1998.0000

Top entities by local observation count.

References

[1] Lori Rosenkopf and Atul Nerkar. Beyond local search: Boundary-spanning, exploration, and impact in the optical disk industry. *Strategic Management Journal*, 22(4):287–306, 2001.